Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Bankman, Issac Ph.D.		
Project Title:	Space Qualifiable Magnetic Resonance Imaging System		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	isaac.bankman@jhuapl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	443-778-6097
Organization Name:	The Johns Hopkins University		
PI Address 1:	Applied Physics Laboratory		
PI Address 2:	11100 Johns Hopkins Road		
PI Web Page:			
City:	Laurel	**State:**	MD
Zip Code:	20723	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	07/01/2002	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Paul Bottomley		
COI Name (Institution):			
Grant/Contract No.:	TD.002.02		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Magnetic resonance imaging (MRI) could serve as the primary tool for non-invasively quantifying and monitoring anatomic and functional changes associated with low gravity in a controlled space environment in longitudinal studies, and permit the quantitative assessment of interventions designed to counteract such changes in animal models. Thus, the goal of this project is to build a space-qualifiable prototype MRI system with an imaging aperture of 5-6 cm. Target weight and power requirements for the initial version are <150 kg, and < 1.5 kW respectively. We plan to base the system on a commercial MRI spectrometer, which will greatly reduce costs associated with sophisticated MRI pulse sequence programming, and the development of high performance RF and interface hard-ware. This will shift the focus of the effort to efficient power and weight design. Magnet manufacturers are working on HTS magnets, and we believe that this solution will be available in the future to provide a light-weight magnet. Since many of the changes caused by low gravity involve anatomical and functional changes affecting internal organs, magnetic resonance imaging (MRI)		

	would provide a primary yard-stick for non-invasively quantifying anatomic and physiological parameters in a real, controlled, space environment in longitudinal studies, and permit the quantitative assessment of interventions designed to counteract these changes in animal models. Thus, the goal of this project is to build a space qualifiable proto-type MRI system with an imaging aperture suitable for imaging rodents in animal models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since many of the changes caused by low gravity involve anatomical and functional changes affecting internal organs, magnetic resonance imaging (MRI) would provide a primary yard-stick for non-invasively quantifying anatomic and physiological parameters in a real, controlled, space environment in longitudinal studies, and permit the quantitative assessment of interventions designed to counteract these changes in animal models. Thus, the goal of this project is to build a space qualifiable proto-type MRI system with an imaging aperture suitable for imaging rodents in animal models.
Task Progress:	No progress reported this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)